

Real-time Cooperative Spectrum Sensing: Reliable Localization of Unknown Interferers in Shared Spectrum

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Abstract—Locating active users/transmitters and sources of spectral interference in real-time is a key challenge in next generation radio networks to enable robust wireless communication and highly dynamic radio resource allocation. We present and investigate a spectrum sensing network with multiple collaborating spectrum sensing units (SUs) which jointly find precise locations of active wireless transmitters in an observed area. The estimation of locations is based on measurements of received signal strength at SUs, taking into account that the measurements are perturbed by spatially correlated shadowing. We propose four localization methods for the cases of single and multiple active transmitters, namely, maximum likelihood search, generalized least squares, radio environment map (REM) based and compressed sensing based methods. The methods vary in their estimation accuracy and complexity. Simulation results present effects of various parameters on the localization performance and also show that there is no significant performance loss due to discretization.

Index Terms—Spectrum sensing network, maximum likelihood, localization, radio environment map

I. INTRODUCTION

Next generation wireless networks, e.g. 5G cellular networks, are envisioned with four major objectives: low latency, higher reliability, high system capacity and massive device connectivity. More spectrum is needed to achieve these objectives. For this purpose, techniques such as license-shared-access and license-assisted-access have been proposed to aggregate spectrum of licensed and unlicensed bands. However, communication in shared spectrum comes with its own challenges—interference and coexistence. The desire for reliable communication in dynamically varying environments such as industrial communication necessitates an intelligent radio access network that is able to sensitively monitor the spectrum, to detect sources of disturbances and to adapt itself accordingly. To enable spectrum and location awareness, spectrum sensing network (SSN) is a potential direction to explore. Analogous to the cognitive radio networks, a spectrum sensing network consists of sensing units (SU¹) distributed in the area. SUs sense spectrum to find frequency bands that are temporally unoccupied by transmitters (TXs), the so-called spectrum holes. However, in SSN, spectrum sensing is not just limited to finding spectrum holes but can also be used to determine transmitter locations. TXs may be users in coexisting networks or jammers trying to disrupt the communication. SSN can also construct a radio environment

map (REM) of the observed area and enable the communication network to make intelligent decisions regarding radio resource management (RRM) within this area [1]. This paper focuses on localization of unknown TXs with multiple SUs collecting and fusing information of spectrum usage at different positions.

Localization methods are broadly categorized as angle-of-arrival (AoA), time-of-arrival (ToA), time-difference-of-arrival (TDoA) and received-signal-strength (RSS) based [2], [3]. First three categories require transmission of known signals between localizing sensors and the object that has to be localized. As signal from an unknown TX is not known a-priori, AoA, ToA and TDoA methods are usually not suitable. However, RSS based methods can be used since localization is done solely based on received power. Least squares (LS) estimation is commonly used in RSS based methods [3]–[8]. In these methods, a TX is localized using its distance from all SUs, which depends on a known path-loss model. However, in practice, measurements at SUs differ from actual received power due to shadowing. The works mentioned above do not consider shadowing. Yilmaz et. al. [8] considered that shadowing-effect is eliminated by taking mean of a large sample of RSS values. In [9], authors used semi-definite programming to provide solution to the non-convex ML formulation. However, these methods consider that only one TX is present. Guibene et. al. [10] considered multiple active TXs and used a basis pursuit algorithm to estimate TX locations without considering shadowing.

In indoor industrial scenarios, it is important to consider shadowing due to scattered obstacles in the area. Further, the shadowing is correlated, that is, SU measurements shadowed by the same obstacle undergo similar perturbations. Thus, in this paper, we consider that RSS measurements at SUs are affected by spatially correlated shadowing. We use discrete coordinate space, obtained by dividing the observed area in a grid of pixels. This enables formulation of localization problems in reduced search space. Results show that grid resolution does not affect the performance significantly. The main contributions of this paper are the following:

- 1) In a correlated shadowing environment, we consider two cases: single active TX and multiple active TXs. For single TX case, we propose two localization methods: maximum-likelihood-search (MLS) and generalized-least-squares (GLS).
- 2) We propose two methods for the case of multiple TXs, namely, radio environment map localization (REML) and compressed-sensing (CS) based.

¹In cognitive radio terminology, the abbreviation SU indicates a secondary user. But in context of spectrum sensing, the terms secondary user and sensing unit are interchangeable.

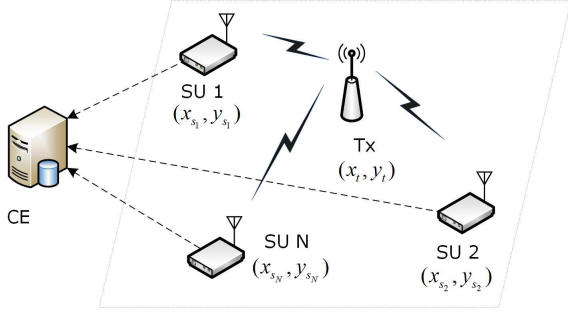


Fig. 1. Scenario with transmitters and sensing units distributed in an area

- 3) Finally, we provide a comparison of all methods in terms of their localization performance and time complexity through simulations, considering pathloss model in industrial scenario. We investigate the effect of number of sensors, number of transmitters and grid resolution on the performance.

II. SYSTEM MODEL

The system model is as shown in Fig. 1. We consider that M TXs with omni-directional antenna and N SUs are distributed in an area of dimensions $L \times L$ m². SUs measure received power in a given frequency band at their respective locations and communicate the RSS measurements to a cognitive engine (CE). We assume that the backhaul links between SUs and CE are error-free, synchronized and not backlogged. CE has the knowledge of SU locations. Using the measurements received from SUs, CE localizes active TXs in the area.

We denote the location of i th TX as tuple $\mathbf{u}_{t_i} := (x_{t_i}, y_{t_i})$ and the location of j th SU as tuple $\mathbf{u}_{s_j} := (x_{s_j}, y_{s_j})$. Then the distance between two SUs is $\|\mathbf{u}_{s_i} - \mathbf{u}_{s_j}\| = \sqrt{(x_{s_i} - x_{s_j})^2 + (y_{s_i} - y_{s_j})^2}$, where $\|\cdot\|$ denotes ℓ_2 -norm. As done in [1], we assume that effect of small-scale fading is eliminated owing to long-term measurements. In logarithmic domain², power received by j th SU is given by [11]

$$z_j = \ln \left(\sum_{i=1}^M p_{t_{x,i}} \left(\frac{\|\mathbf{u}_{t_i} - \mathbf{u}_{s_j}\|}{d_{\text{ref}}} \right)^{-\gamma} \right) + w_j, \quad (1)$$

where $p_{t_{x,i}}$ is the transmit power of i th TX, w_j is the shadowing gain, d_{ref} is reference distance and γ is path-loss exponent of the propagation model. We consider log-normal shadowing with standard deviation σ . Then for $\mathbf{w} = [w_1, w_2, \dots, w_N]^T$, we have $\mathbf{w} \sim \mathcal{N}(\mathbf{0}_{(N \times 1)}, \mathbf{\Sigma}_{(N \times N)})$. Shadowing follows distance based correlation [12]. Thus, we can write $\Sigma_{ij} = \sigma^2 \exp \left(\frac{-\|\mathbf{u}_{s_i} - \mathbf{u}_{s_j}\|}{d_{\text{corr}}} \right)$, where d_{corr} is the correlation distance. We assume that path-loss exponent γ and covariance matrix $\mathbf{\Sigma}$ are known to CBS owing to measurement studies.

Three localization methods (MLS, REML, CS) proposed in this paper work in discrete coordinate space. For this purpose, we divide the area in a pixel-grid with resolution n_{res} pixels/meter. Each side of the area contains L_{pixel} pixels. Total number of pixels

in the area is denoted by $K = L_{\text{pixel}}^2$. The transformation from continuous space coordinates to pixel coordinates is given by $(x_{\text{pixel}}, y_{\text{pixel}}) = \Psi(x, y)$, where the transformation function Ψ is defined as $\Psi(x) = \lceil n_{\text{res}} x \rceil + 1$. The inverse transformation is defined as $\Psi^{-1}(x_{\text{pixel}}) = \left\lfloor \frac{x_{\text{pixel}} - 1}{n_{\text{res}}} \right\rfloor$.

III. PROPOSED LOCALIZATION METHODS

We propose four localization methods of varying localization performance and complexity. MLS and GLS assume that only one TX is present while REML and CS work even when multiple TXs are present. We address these two cases separately in this section.

A. Single transmitter present

In this case, TX location is denoted simply as $\mathbf{u}_t := (x_t, y_t)$.

1) *Maximum likelihood search (MLS)*: This method chooses the location that maximizes a-posteriori probability of receiving specific power levels at SUs for given TX location. RSS at j th SU is given by

$$z_j = \mu_j + w_j, \quad (2)$$

where $\mu_j = \ln(p_{t_{x}}) + \gamma \ln d_{\text{ref}} - \gamma \ln(\|\mathbf{u}_t - \mathbf{u}_{s_j}\|)$. Let $\boldsymbol{\mu} = [\mu_1, \mu_2, \dots, \mu_N]$. Then joint probability of receiving $\mathbf{z} = [z_1, z_2, \dots, z_N]^T$ for given TX location $\mathbf{u}_t$ is given by

$$f(\mathbf{z}|\mathbf{u}_t) = \frac{1}{(2\pi)^{N/2} |\mathbf{\Sigma}|^{\frac{1}{2}}} \exp \left(-\frac{1}{2} (\mathbf{z} - \boldsymbol{\mu})^T \mathbf{\Sigma}^{-1} (\mathbf{z} - \boldsymbol{\mu}) \right). \quad (3)$$

Estimated TX location is the coordinate $\hat{\mathbf{u}}_t$ that maximizes a-posteriori probability in (3). Thus, we can write

$$\hat{\mathbf{u}}_t := \arg \min_{(x_t, y_t)} (\mathbf{z} - \boldsymbol{\mu})^T \mathbf{\Sigma}^{-1} (\mathbf{z} - \boldsymbol{\mu}). \quad (4)$$

Since $\ln f(\mathbf{z}|\mathbf{u}_t)$ is a non-convex function of x_t and y_t , optimization method like gradient descent is unable to find the global minima. However, exhaustive search in a discrete coordinate space can find pixel location that maximizes the likelihood. Thus, we create a matrix $\mathbf{U}$ of size $(L_{\text{pixel}} \times L_{\text{pixel}})$ where each element is the log-likelihood value in (3) for corresponding pixel location. Then TX location is $(\hat{x}_t, \hat{y}_t) = \Psi^{-1}(\hat{x}_{\text{pixel}}, \hat{y}_{\text{pixel}})$, where

$$(\hat{x}_{\text{pixel}}, \hat{y}_{\text{pixel}}) := \arg \min_{(i,j)} U_{ij}.$$

MLS finds TX location with complexity $\mathcal{O}(KN)$, where N is the number of sensors and K is the number of pixels in the area.

2) *Generalized least squares (GLS)*: Rearranging (2) and using (1), we can write

$$\|\mathbf{u}_t - \mathbf{u}_{s_j}\| = \exp \left(\frac{\ln p_{t_{x}} + \gamma \ln d_{\text{ref}} - z_j}{\gamma} + \frac{w_j}{\gamma} \right). \quad (5)$$

Further simplifying, we get

$$x_{s_j}^2 + y_{s_j}^2 = 2x_t x_{s_j} + 2y_t y_{s_j} - x_t^2 - y_t^2 + \exp \left(\frac{2 \ln p_{t_{x}} + 2\gamma \ln d_{\text{ref}} - 2z_j}{\gamma} + \frac{2w_j}{\gamma} \right). \quad (6)$$

For N SUs, we have a system of linear equations written as

$$\mathbf{b}_{(N \times 1)} = \mathbf{A}_{(N \times 3)} \mathbf{x}_{(3 \times 1)} + \mathbf{n}_{(N \times 1)}, \quad (7)$$

²In decibels, $z_{(\text{dB})} = z_{(\text{Np})} \times 10 \log_{10} e$.

where

$$\mathbf{x} = \begin{bmatrix} x_t \\ y_t \\ x_t^2 + y_t^2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} x_1^2 + y_1^2 \\ x_2^2 + y_2^2 \\ \vdots \\ x_N^2 + y_N^2 \end{bmatrix}, \mathbf{A} = \begin{bmatrix} 2x_1 & 2y_1 & -1 \\ 2x_2 & 2y_2 & -1 \\ & & \ddots \\ 2x_N & 2y_N & -1 \end{bmatrix},$$

$$\text{and } \mathbf{n} = \begin{bmatrix} \exp\left(\frac{2 \ln p_{tx} + 2\gamma \ln d_{ref} - 2z_1}{\gamma} + \frac{2w_1}{\gamma}\right) \\ \exp\left(\frac{2 \ln p_{tx} + 2\gamma \ln d_{ref} - 2z_2}{\gamma} + \frac{2w_2}{\gamma}\right) \\ \vdots \\ \exp\left(\frac{2 \ln p_{tx} + 2\gamma \ln d_{ref} - 2z_N}{\gamma} + \frac{2w_N}{\gamma}\right) \end{bmatrix}.$$

The unknown $\mathbf{x}$ contains the TX location. As $\mathbf{w}$ is a vector of correlated Gaussian random variables, $\mathbf{n}$ is a vector of correlated log-normal random variables with mean μ_{ls} and covariance matrix Σ_{ls} . We can write

$$\mu_{ls,j} = \exp\left(\frac{2 \ln p_{tx} + 2\gamma \ln d_{ref} - 2z_j}{\gamma} + \frac{2\Sigma_{jj}}{\gamma^2}\right), \quad (8)$$

$$\Sigma_{ls,ij} = \exp\left(\frac{4 \ln p_{tx} + 4\gamma \ln d_{ref} - 2(z_i + z_j)}{\gamma} + \frac{2(\Sigma_{ii} + \Sigma_{jj})}{\gamma^2}\right) \times (\exp(\Sigma_{ij}) - 1). \quad (9)$$

The least squares objective is to find $\mathbf{x}$ that minimizes Mahalanobis distance between $\mathbf{b}$ and $\mathbf{A} \cdot \mathbf{x}$ under the constraint $x_1^2 + x_2^2 = x_3$. Thus, the optimization problem is

$$\min_{\mathbf{x}} (\mathbf{b} - \mathbf{A}\mathbf{x} - \mu_{ls})^T \Sigma_{ls}^{-1} (\mathbf{b} - \mathbf{A}\mathbf{x} - \mu_{ls}) \quad (10a)$$

$$\text{s. t. } \mathbf{r}^T \mathbf{x} + \mathbf{x}^T \mathbf{Q} \mathbf{x} = 0, \quad (10b)$$

where

$$\mathbf{r} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \text{ and } \mathbf{Q} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

The problem is solved using Lagrangian method. We write the Lagrangian as

$$\mathcal{L}(\mathbf{x}, \lambda) = (\mathbf{b} - \mathbf{A}\mathbf{x} - \mu_{ls})^T \Sigma_{ls}^{-1} (\mathbf{b} - \mathbf{A}\mathbf{x} - \mu_{ls}) + \lambda (\mathbf{r}^T \mathbf{x} + \mathbf{x}^T \mathbf{Q} \mathbf{x}), \quad (11)$$

where λ is Lagrange's multiplier. Differentiating with respect to $\mathbf{x}$ and λ and equating to zero, we get

$$\frac{\partial \mathcal{L}(\mathbf{x}, \lambda)}{\partial \mathbf{x}} = 2\mathbf{x}^T (\mathbf{A} \Sigma_{ls}^{-1} \mathbf{A} + \lambda \mathbf{Q}) - 2(\mathbf{b} - \mu_{ls})^T \Sigma_{ls}^{-1} \mathbf{A} + \lambda \mathbf{r}^T = 0, \quad (12)$$

$$\frac{\partial \mathcal{L}(\mathbf{x}, \lambda)}{\partial \lambda} = \mathbf{r}^T \mathbf{x} + \mathbf{x}^T \mathbf{Q} \mathbf{x} = 0. \quad (13)$$

From (12), we get the solution in terms of λ as

$$\mathbf{x} = (\mathbf{A} \Sigma_{ls}^{-1} \mathbf{A} + \lambda \mathbf{Q})^{-1} \left(\mathbf{A}^T \Sigma_{ls}^{-1} (\mathbf{b} - \mu_{ls}) - \frac{\lambda \mathbf{r}}{2} \right). \quad (14)$$

We can find optimal λ by substituting (14) in (13) and using standard root finding methods, e.g. bisection search. Complexity of GLS method is $\mathcal{O}(N \log N)$.

Note that, when underlying distribution in measurements is Gaussian, least-square and maximum-likelihood methods are the same. However, in localization problem, the observation vector is perturbed by a lognormal distribution as seen from (7). Hence MLS and GLS are two different localization methods.

B. Multiple transmitters present

1) *Radio environment map localization (REML)*: In this method, we first estimate a REM that indicates received power level at each pixel in the area. Then, we process the REM to extract pixel locations with higher estimated power than the rest, which are potential TX locations. REM is constructed using ordinary Kriging interpolation as it does not require knowledge of the mean power at each location in the area [1]. Kriging works in two steps: first, we determine spatial-covariance structure of received power from measurements (represented by semivariogram). Then, we estimate received power at an arbitrary point using a minimum-mean-squared-error linear estimator. Let $\mathbf{z} = [z_1, z_2, \dots, z_N]$ denote the measurement vector at SUs. We first calculate semivariogram $\gamma(d_{ij}) = \frac{1}{2} \text{Var}(z_i - z_j)$, where $d_{ij} = \|\mathbf{u}_{s_i} - \mathbf{u}_{s_j}\|$. Let $\mathcal{S}(d)$ denote set of all SUs within distance d from each other. Then, semivariogram is estimated as [1]

$$\hat{\gamma}(d) \approx \frac{1}{2|\mathcal{S}(d)|} \sum_{i,j \in \mathcal{S}(d)} (z_i - z_j)^2, \quad (15)$$

where $|\mathcal{S}(d)|$ denotes cardinality of the set. The empirical semi-variogram is then fitted to an exponential model given by

$$\gamma(d_{ij}) = \alpha_n^2 + \alpha_s^2 \left[1 - \exp\left(-\frac{d_{ij}}{\alpha_r}\right) \right], \quad (16)$$

where parameters α_n , α_s and α_r are obtained from least-squares fit. Kriging method estimates received power z_0 at an arbitrary location $\mathbf{u}_0$ using a linear estimator $z_0 = \sum_{i=1}^N v_i z_i$ where z_i is the measurement at i th SU. Weights v_i , $i = 1, \dots, N$ are chosen so as to minimize estimation error $\mathbb{E}[(z_0 - z_0)^2]$ and satisfy unbiasedness condition $\sum_{i=1}^N v_i = 1$. The optimization problem is written as [1]

$$\min_{\mathbf{v}} -\gamma(d_{00}) - \sum_{i=1}^N \sum_{j=1}^N v_i v_j \gamma(d_{ij}) + 2 \sum_{i=1}^N v_i \gamma(d_{i0}) \quad (17a)$$

$$\text{s. t. } \sum_{i=1}^N v_i = 1. \quad (17b)$$

Associating Lagrange's multiplier μ with constraint (17b) and using Lagrange's method, we get

$$\begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_N \\ \mu \end{bmatrix} = \begin{bmatrix} \gamma(d_{11}) & \cdots & \gamma(d_{1N}) & 1 \\ \gamma(d_{21}) & \cdots & \gamma(d_{2N}) & 1 \\ \vdots & \vdots & \vdots & \vdots \\ \gamma(d_{N1}) & \cdots & \gamma(d_{NN}) & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}^{-1} \begin{bmatrix} \gamma(d_{10}) \\ \gamma(d_{20}) \\ \vdots \\ \gamma(d_{N0}) \\ 1 \end{bmatrix}. \quad (18)$$

To obtain REM, Kriging is performed for each pixel in the area. Let $\mathbf{U}_{(L_{\text{pixel}} \times L_{\text{pixel}})}$ denote the REM. For single-TX case, the pixel coordinates of the transmitter are

$$(x_{\text{pixel}}, y_{\text{pixel}}) := \arg \max_{(i, j)} U_{ij}. \quad (19)$$

Thus, complexity of REML method is $\mathcal{O}(KN \log N)$. For multiple-TXs case, we perform histogram thresholding [13] on $\mathbf{U}$, followed by image segmentation to get high-power regions. Pixel locations with the highest value in each region are then considered as the TX locations.

2) *Compressed sensing (CS)*: The methods discussed above assume that all TXs transmit with the same power and CE has the knowledge of transmit power. CS method does not need these assumptions. Consider a matrix $\mathbf{U}_{(L_{\text{pixel}} \times L_{\text{pixel}})}$ where U_{ij} is transmit power of a potential TX located at pixel location (i, j) . Thus, $\mathbf{U}$ is a sparse matrix containing transmit powers where positions of non-zero elements indicate TX locations. We can write

$$U_{ij} = \begin{cases} p_{ij} & \text{for } (i, j) \in \{\Psi(\mathbf{u}_{t_1}), \dots, \Psi(\mathbf{u}_{t_M})\} \\ 0 & \text{otherwise,} \end{cases} \quad (20)$$

where p_{ij} is the transmit power of TX at location (i, j) . Similarly, let $\mathbf{C}_{(L_{\text{pixel}} \times L_{\text{pixel}})}^{(j)}$ be the matrix of path-loss factors between all pixel locations and j th SU. That is, $C_{ij}^{(k)} = \left(\frac{\|\Psi^{-1}(i, j) - \mathbf{u}_{s_k}\|}{d_{\text{ref}}} \right)^{-\gamma}$. Then total received power at k th SU, without considering shadowing, is given by $\sum_{i=1}^{L_{\text{pixel}}} \sum_{j=1}^{L_{\text{pixel}}} C_{ij}^{(k)} U_{ij}$. For the case of shadowing, from (1), we can write

$$\exp(z_k - w_k) = \sum_{i=1}^{L_{\text{pixel}}} \sum_{j=1}^{L_{\text{pixel}}} C_{ij}^{(k)} U_{ij}. \quad (21)$$

We unroll matrix $\mathbf{U}$ to get a sparse vector representing transmit powers at all pixel locations, denoted as $\mathbf{x}_{(L_{\text{pixel}}^2 \times 1)}$. Note that $\mathbf{x}$ is the unknown that contains information of TX locations and their transmit powers. Similarly, we unroll $\mathbf{C}^{(k)}$ to get vector $\mathbf{a}_{(L_{\text{pixel}}^2 \times 1)}^{(k)}$. Thus, we have $z_k = (\mathbf{a}^{(k)})^T \mathbf{x} + w_k$. Then, for measurements at N SUs, we can write

$$\mathbf{n}_{(N \times 1)} = \mathbf{A}_{(N \times L_{\text{pixel}}^2)} \mathbf{x}_{(L_{\text{pixel}}^2 \times 1)}, \quad (22)$$

where

$$\mathbf{n} = \begin{bmatrix} \exp(z_1 - w_1) \\ \exp(z_2 - w_2) \\ \vdots \\ \exp(z_N - w_N) \end{bmatrix} \text{ and } \mathbf{A} = \begin{bmatrix} (\mathbf{a}^{(1)})^T \\ (\mathbf{a}^{(2)})^T \\ \vdots \\ (\mathbf{a}^{(N)})^T \end{bmatrix}.$$

Vector $\mathbf{n}$ is a vector of correlated log-normal random variables with mean $\boldsymbol{\mu}_{\text{cs}}$ and covariance matrix $\boldsymbol{\Sigma}_{\text{cs}}$. Similar to GLS method, we have

$$\mu_{\text{cs},j} = \exp\left(z_j + \frac{\Sigma_{jj}}{2}\right), \quad (23)$$

$$\Sigma_{\text{cs},ij} = \exp\left(z_i + z_j + \frac{\Sigma_{ii} + \Sigma_{jj}}{2}\right) (\exp(\Sigma_{ij}) - 1). \quad (24)$$

The objective is to find sparse vector $\mathbf{x}$ that minimizes Mahalanobis distance between LHS and RHS in (22). Similar to GLS method, we can write the optimization problem as

$$\min_{\mathbf{x}} \quad (\mathbf{A}\mathbf{x} - \boldsymbol{\mu}_{\text{cs}})^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} (\mathbf{A}\mathbf{x} - \boldsymbol{\mu}_{\text{cs}}) \quad (25a)$$

$$\text{s. t.} \quad \|\mathbf{x}\|_0 \leq \beta. \quad (25b)$$

The constraint in (25b) is sparsity constraint and β is the limit on number of non-zero entries in $\mathbf{x}$. Sparse vector recovery with ℓ_0 -norm is a NP-hard problem [14]. Thus, we simplify the problem by considering ℓ_1 -norm, as done in [10]. Then the optimization problem is written as

$$\min_{\mathbf{x}} \quad (\mathbf{A}\mathbf{x} - \boldsymbol{\mu}_{\text{cs}})^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} (\mathbf{A}\mathbf{x} - \boldsymbol{\mu}_{\text{cs}}) \quad (26a)$$

$$\text{s. t.} \quad \|\mathbf{x}\|_1 = \mathbf{r}^T \mathbf{x} \leq \beta, \quad (26b)$$

$$x_i \geq 0, i = 1, \dots, N, \quad (26c)$$

where $\mathbf{r}_{(N \times 1)} = [1, \dots, 1]^T$.

We first ignore the non-negativity constraint (26c) and solve the problem in (26a) using Lagrange's method. Lagrangian is written as

$$\mathcal{L}(\mathbf{x}, \lambda) = (\mathbf{A}\mathbf{x} - \boldsymbol{\mu}_{\text{cs}})^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} (\mathbf{A}\mathbf{x} - \boldsymbol{\mu}_{\text{cs}}) + \lambda (\mathbf{r}^T \mathbf{x} - \beta), \quad (27)$$

where λ is Lagrange's multiplier. Differentiating (27) with respect to $\mathbf{x}$ and λ , we get

$$\frac{\partial \mathcal{L}}{\partial \mathbf{x}} = 2\mathbf{x}^T \mathbf{A}^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} \mathbf{A} - 2\boldsymbol{\mu}_{\text{cs}}^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} \mathbf{A} + \lambda \mathbf{r}^T = 0, \quad (28)$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = \mathbf{r}^T \mathbf{x} - \beta = 0. \quad (29)$$

Equating (28) and (29) with zero, we get the solution as

$$\mathbf{x} = (\mathbf{A}^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} \mathbf{A})^{-1} \left(\mathbf{A}^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} \boldsymbol{\mu}_{\text{cs}} - \frac{\lambda \mathbf{r}}{2} \right), \quad (30)$$

where λ is given by

$$\lambda = 2 \left[\mathbf{r}^T (\mathbf{A}^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} \mathbf{A})^{-1} \mathbf{r} \right]^{-1} \times \left[\mathbf{r}^T (\mathbf{A}^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} \mathbf{A})^{-1} \mathbf{A}^T \boldsymbol{\Sigma}_{\text{cs}}^{-1} \boldsymbol{\mu}_{\text{cs}} - \beta \right]. \quad (31)$$

To ensure that $\mathbf{x}$ has non-negative values, we use an iterative approach. In each iteration, we define a set of indices $\mathcal{S} := \{i | x_i < 0\}$. As powers cannot be negative, in the next iteration, locations $i \in \mathcal{S}$ and the corresponding path-loss vectors $\mathbf{a}^{(i)}$ are not considered. Thus, in the new optimization problem, $\mathbf{A}$ has dimension $(N \times (L_{\text{pixel}}^2 - 1))$ and $\mathbf{x}$ has dimension $((L_{\text{pixel}}^2 - 1) \times 1)$. The optimization problem in (26a) is solved repeatedly until $\mathbf{x}$ is non-negative. Detailed algorithm is presented in Algorithm 1. CS method estimates transmitter locations with complexity $\mathcal{O}(KN \log K)$.

Algorithm 1 Algorithm to ensure non-negative x_i 's in CS method

- 1) while $|\mathcal{S}| > 0$
- 2) Find $\mathbf{x}^{(n)}$ using (30) and (31), where n is iteration index.
- 3) Update $\mathcal{S} := \{i \mid x_i < 0\}$.
- 4) Update $\mathbf{x}$ and $\mathbf{A}$ such that

$$\begin{aligned}\{\mathbf{x}^{(n+1)}\} &:= \{\mathbf{x}^{(n)}\} - \{x_{i \in \mathcal{S}}\} \\ \{\mathbf{A}^{(n+1)}\} &:= \{\mathbf{A}^{(n)}\} - \{\mathbf{a}^{(i \in \mathcal{S})}\},\end{aligned}$$

where $\{\mathbf{x}\}$ is a set of all elements x_i and $\{\mathbf{A}\}$ is the set of all rows $\mathbf{a}^{(i)}$.

- 5) Repeat Step 2.
 - 6) end while
-

TABLE I
COMPARISON OF PROPOSED LOCALIZATION METHODS

Method	Assumption	Space	#Tx	Complexity
MLS	Known p_{tx}	Discrete	= 1	$\mathcal{O}(KN)$
GLS	Known p_{tx}	Continuous	= 1	$\mathcal{O}(N \log N)$
REML	All $p_{tx,i}, \forall i$ equal	Discrete	≥ 1	$\mathcal{O}(KN \log N)$
CS	-	Discrete	≥ 1	$\mathcal{O}(KN \log K)$

Assumptions and limitations: We now briefly discuss salient points of all methods. Three localization methods—MLS, REML and CS—work with discrete coordinate space, that is, the area is divided into a grid of pixels. This is not required for GLS method. MLS and GLS assume that only one TX is present, whereas REML and CS methods can localize multiple TXs. REML takes an image processing approach where the REM is segmented to find regions of high power. For this purpose, REML assumes that all TXs transmit with the same power. CS method makes no such assumption about transmit powers. CS estimates locations of multiple TXs as well as their transmit powers. Finally, variogram-fitting in REML and the number of unknowns in other methods dictate that at least 3 measurements are needed to perform localization, i.e. $N \geq 3$. This comparison is summarized in Table I.

IV. SIMULATION RESULTS AND DISCUSSION

For simulation, we consider a square shaped area with dimension $L = 20$ m. TXs are equidistantly situated on the horizontal line passing through the centre with a separation of 5 m. SUs are distributed randomly (with uniform distribution) in a circle of radius $R = 10$ m at the centre, unless mentioned otherwise. All TXs have the same transmit power $p_{tx} = 0.1$ W. We use the path-loss model for indoor industrial channel proposed in [15] with path-loss exponent $\gamma = 2.16$, $d_{ref} = 1$ m and shadowing deviation $\sigma = 8$ dB. The correlation distance is $d_{corr} = 5$ m. Pixel resolution is $n_{res} = 2$ pixels/m, unless mentioned otherwise. We performed 10^4 simulations to evaluate performance of proposed methods.

Two types of errors occur in localization: First, the estimated location does not coincide with the actual TX location; Second,

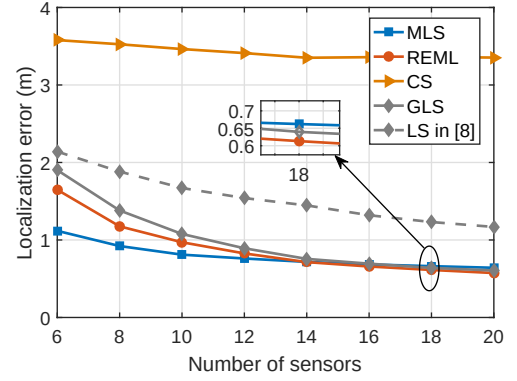


Fig. 2. Effect of number of sensors on localization error in single-transmitter-case.

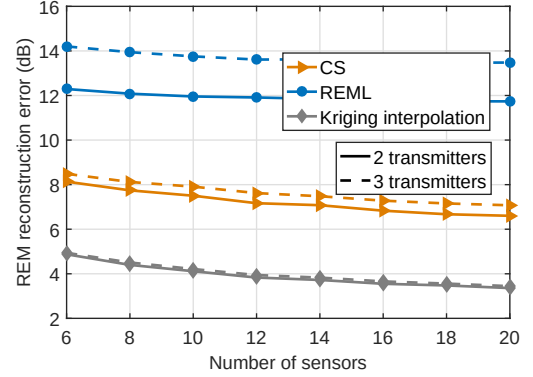


Fig. 3. Effect of number of sensors on REM reconstruction error in multiple-transmitters-case for different number of transmitters.

number of estimated TXs can be more or less than the actual number. Thus, in the case of single TX, we use localization error as the performance metric to compare performance of all localization methods. For the case of multiple TXs, we reconstruct radio environment map from the estimated locations using (1) and the propagation model parameters mentioned above. In this case, the performance metric is root-mean-square reconstruction error (in dB) which is the error between actual power levels in the area and reconstructed power levels. Our focus is on studying estimation performance of proposed methods and not on finding optimal sensor locations.

Localization and reconstruction error: Fig. 2 plots localization error versus number of sensors for the single-TX case. Localization error decreases with increasing N . We see that MLS performs the best when number of SUs is low, followed by REML. When N is high, REML and GLS perform better than MLS. CS has the highest estimation error. We also plot localization error for the least-squares method in [8] which does not consider correlated shadowing. MLS, REML and GLS outperform the method in [8].

Fig. 3 shows reconstruction error in the multiple-TXs case for different number of TXs. Similar to the single-TX case, error decreases with increasing number of SUs. In this case, CS method performs significantly better than REML. Increasing number of TXs results in increasing reconstruction error. However, compared to REML, performance of CS remains relatively unchanged. This is because, with more TXs, high power regions in REM may get merged and the segmentation algorithm in

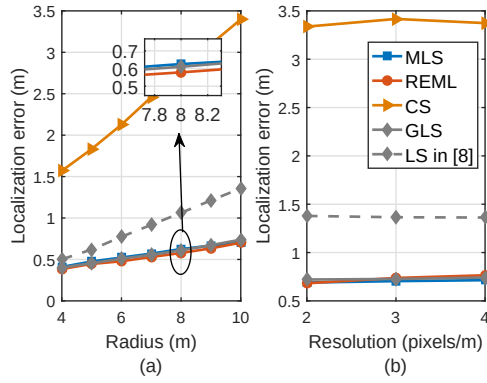


Fig. 4. Effect of varying (a) radius and (b) grid resolution on localization error for $N = 15$ in single-transmitter-case.

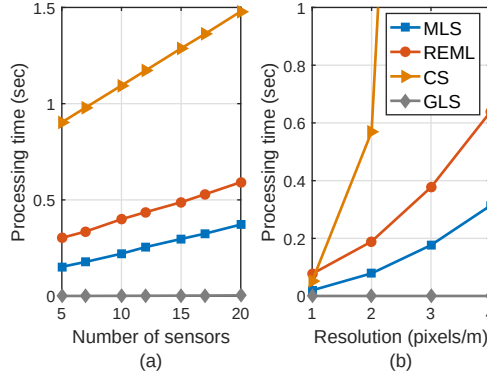


Fig. 5. Effect of increasing (a) number of sensors (for $n_{res} = 2$ pixels/m) and (b) grid resolution (for $N = 10$) on processing time.

REML cannot identify each TX separately. Note that REML itself is based on reconstruction of power levels in the area using Kriging interpolation. For reference, we also plot the error of Kriging interpolation. We can see that reconstruction performance is better than CS. This means that Kriging is good at estimating power levels but not at localization of multiple TXs.

Fig. 4(a) and Fig. 4(b) show the effect of radius R and grid resolution on localization error for single-TX case. Localization error is low for smaller values of R . Thus, localization is better when SUs are located near the TX. As R increases, localization error increases. From Fig. 4(b), we can see that changing grid resolution does not have significant impact on localization performance.

Complexity: Fig. 5(a) and Fig. 5(b) show effect of increasing number of sensors and grid resolution on processing time for all localization methods. Processing time increases with increasing number of SUs. GLS needs the least processing time while processing time for CS is the highest. Increasing pixel resolution means increased search space for the localization methods. We see that, with increasing resolution, processing time increases rapidly for all methods except GLS as GLS method is not based on discrete coordinate space.

V. CONCLUSION AND FUTURE WORK

We proposed four methods to find locations of unknown transmitters using RSS measurements at SUs. When number of SUs is low, MLS performs the best for localization of one

TX. For higher number of sensors, REML outperforms MLS. CS method is suitable in multiple transmitters case. We observe an interesting time-performance tradeoff. Time complexity of CS is the highest, while complexity of GLS is the lowest. Complexity increases rapidly with increasing grid resolution of the area. On the other hand, grid resolution does not affect localization performance significantly. Thus, processing time may be decreased by choosing lower pixel resolution. Future work will focus on the effect of fast fading, time-performance tradeoff and optimal SU deployment.

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